MILITARY SPECIFICATION

GENERATOR TRANSMITTER GROUP
AN/ASA-76 (U)

This specification has been approved by the
Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 Scope - The equipment covered by this specification shall be capable of interrogating and commanding active and passive sonobuoys. It shall provide sonic signals for the active sonobuoys (AN/SSQ-50 - CASS) and (AN/SSQ-62 - DICASS) and shall provide the required mode and control signals to the demultiplexer, digital interface, data processor of the two AN/AQA-7(V) DIFAR systems installed in the P-3 patrol aircraft. The equipment shall provide an interface and be compatible with the aircraft's sonobuoy radio receivers, tape recorder via the sonobuoy junction box and the UHF-2 radio transmitter. No additional wiring or connectors will be required by the equipment other than those specified in Avionics Installation Instruction for Sonar Computer-Recorder Group AQA-7(V), EI-555.

1.2 Classification - The equipment covered by this specification shall consist of the following items:

Item No.	Designation	Applicable Paragraph
1	Control, Reference Signal Generator	3.5.1
2	Reference Signal Generator	3.5.2
3	Radio Transmitter	3.5.3

1.3 Associated Equipment - The items (1.2) shall operate with associated equipment listed in 6.7.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein:

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SPECIFICATIONS

Military

MIL-W-5088	Wiring, Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-C-6781	Control Panel; Aircraft Equipment Rack or Console Mounted
MIL-C-81332	Computer, CP901(V) ASQ-114
MIL-P-7788	Panels, Information, Integrally Illuminated
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and Acceptance for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-T-21303	Thermal Performance Evaluation Airborne Electronic Equipment, General Requirements for
MIL-R-81628	UHF Transceiver, AN/ARC-143
MIL-R-81681	Radio Receiving Set, AN/ARR-75
MIL-S-51349	Sonobuoy AN/SSQ-47
MIL-S-81487	Sonobuoy AN/SSQ-53
MIL-S-81666	Sonobuoy AN/SSQ-50
MIL-A-81486(AS)	Aircraft Data Processor AN/AQA-7(V)

Naval Air Systems Command

AR-5	Microelectronic Devices Used in Avionics Equipment, Procedures for Selection and Approval of

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AR-10	Maintainability of Avionics Equipment and Systems, General Requirements for
AR-34	Failure Classification for Reliability Testing, General Requirements for
WR-101 Part I	Electronic Control Requirements for Advanced ASW Avionics Systems, A-NEW
XAS-1647(V)	Directional Command Active Sonobuoy
XAS-1645	Sonar Display Computer AN/AQA-9

STANDARDS

Military

MIL-STD-415	Test Points and Test Facilities for Electronic Systems and Associated Equipment Design Standard 57
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-470	Maintainability Program Requirements (for Systems and Equipments)
MIL-STD-471	Maintainability Demonstration
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-721	Definition of Terms for Reliability Engineering
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-785	Requirement for Reliability Program for Systems and Equipments
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of

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PUBLICATIONS

Naval Air Systems Command

EI-555 Preliminary Avionics Installation Instruction for Sonar Computer-Recorder Group AN/AQA-7(V)

Preliminary Specification Miniature Sonobuoy Receiver Set AN/ARR-72(U)

Preliminary Draft Program Technical Memorandum for Sonar Computer-Recorder Group AN/AQA-7(V) (P-3C Aircraft)

INDUSTRY STANDARDIZATION DOCUMENTS

USAS Y32.16.1965 Electrical and Electronic Reference Designations

2.1.1 Availability of Documents -

(1) When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Publications and Forms Center, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

(2) Copies XAS-1647(AV), Preliminary Specification Miniature Sonobuoy Receiver Set AN/ARR-72, the Program Technical Memorandum for Sonar Computer-Recorder Group AN/AQA-7(V) and Preliminary Avionics Installation Instruction for Sonar Computer-Recorder Group AN/AQA-7(V) may be obtained upon application to Naval Air Systems Command, Navy Department, Washington, D.C. 20360 (Code AIR-5330D6).

(3) Copies of Industry Standardization Documents may be obtained from United States of America Standards Institute, 10 East 40 Street, New York, New York 10016.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

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(1) Microelectronic technology shall be considered and microelectronic items shall conform to requirements specified herein.

(2) Other parts and materials requirement shall conform to Specification MIL-E-5400.

(3) Nonrepairable subassemblies shall be used in accordance with Specification AR-10 and as outlined in Specification MIL-E-5400.

(4) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials (including electron tubes, transistors and diodes) other than microelectronic devices shall be obtained as outlined in Specification MIL-E-5400. Microelectronic devices shall be approved as outlined in AR-5.

3.2.2 Microelectronic Modular Assemblies - When used, Microelectronic Modular Assemblies shall meet the requirements of Specification AR-5. Conformal coatings, encapsulants, embedments or potting materials used with modular assemblies containing integrated circuits and discrete parts shall be easily removable without damage to the assembly.

3.2.3 Modules - The electronic portions of the equipment shall be modularized in accordance with Specification AR-10 as amended herein.

3.3 Design and Construction - The equipment shall fulfill all applicable design, construction and workmanship requirements of Specification MIL-E-5400 unless otherwise specified herein.

3.3.1 Performance objectives - Minimum weight and size, simplicity of operation, ease of maintenance, and improvement in the performance and reliability of specific functions are the objectives in the production of the equipment specified herein. The equipment shall perform as required herein when integrated with the AN/AQA-7(V).

3.3.1.1 Total Weight - The total weight of the equipment, excluding cable but including mounts, shall be a minimum consistent with good design and shall not exceed 65 pounds.

3.3.2 Reliability - The reliability program shall be in accordance with MIL-STD-785.

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3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 300 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The equipment shall have a total operating life of 30,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.3.2.3 Reliability in MTBF (Mean Time Between Failure) - The equipment, including built-in-test features, shall have 300 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.3.

3.3.3 Cabling and Connections -

3.3.3.1 Cables and Connectors - The equipment shall use cables and connectors in accordance with Specification MIL-E-5400.

3.3.3.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.3.4 Control Panels - All rack or console mounted control panels shall conform to the applicable requirements of Specification MIL-C-6781 except as modified by paragraph 3.5.1 of this specification. The configuration of all control panels will be approved by the procuring activity prior to preproduction testing.

3.3.5 Interchangeability - The equipment shall meet the interchangeability requirements of Specification MIL-E-5400.

3.3.6 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification WR-101 Part I and MIL-STD-461.

3.3.6.1 Compatibility - It is necessary that this equipment operate satisfactorily in a military environment containing many types of electronic equipment. In order to insure compatible operation in this environment, it is necessary that the equipment be made free of spurious and unnecessary radiation and not susceptible to the reception of unwanted or unneeded energy.

3.3.7 Provisions for Maintainability - Built-in test equipment, construction and packaging, and provisions for test points shall be as specified

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herein and generally in accordance with Specification AR-10, as amended herein.

3.3.7.1 Maintainability Program - The contractor shall conduct a maintainability program in accordance with MIL-STD-470.

3.3.7.2 Maintainability Requirements - The equipment shall be such that all corrective maintenance can be accomplished by maintenance personnel who have attained the E-4 skill level and who have successfully completed appropriate training programs.

3.3.7.3 Organizational Maintainability - The characteristics of the equipment shall be such that it will be possible in 95% of the cases of failure, to perform all corrective organization maintenance actions, other than combat damage repair, within a period not exceeding 30 minutes maximum (the 30 minute corrective action time shall include all elements except aircraft access time). As here applied, an organizational corrective maintenance action includes the following:

(1) Verification of a fault
(2) Location of the fault to a WRU (Weapon Replaceable Unit)
(3) Correction or repair
(4) Adjustment or alignment (when required)
(5) Checkout of the repair

3.3.7.3.1 Maintenance Concept -

(a) By utilization of BITE and on-board test equipment faults will be isolated to the appropriate module or replaceable assembly then removed and replaced.

(b) Defective units 1 or 3 will be limited to utilizing BITE and on-board test equipment to fault isolate to a defective unit. The faulty unit will be repaired at either the intermediate or depot maintenance level.

(c) Maintenance adjustments shall be capable of being made in the aircraft using simple hand tools and/or general and standard test equipment. Adjustments shall be identified and accessible without a requirement to remove or relocate the module or any other module or part.

3.3.7.3.2 Failure Detection Indicator - The system fault indicator for the BIT function shall be integrated into the control surface of the unit 1. Certain faults will be recognized and isolated by means of the AN/AQA-7(V) operational displays. (See also 3.4.8.)

3.3.7.4 BIT (Built-In-Test) Function - Suitable BIT features shall be incorporated into the equipment to provide both a failure detection function

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and a failure location function.

3.3.7.4.1 BIT (Built-In-Test) Operation - The operation of BIT shall be both computer and manually initiated with provisions for exercising all functions of the equipment to obtain a GO/NO-GO indication, and for isolating faults to a WRU (Weapon's Replaceable Unit).

3.3.7.4.2 BIT (Built-In-Test) Indicators - The BIT function shall provide the equipment operator with an indication of equipment status by energizing an advisory or caution indicator when equipment performance is below an acceptable level, both while on the ground and in flight.

3.3.7.4.2.1 BIT (Built-In-Test) WRU Indicators - Each WRU shall contain a warning indicator which is energized when the unit operation falls below specified levels. The indicator shall be on the surface of the WRU and visible to the operator.

3.3.7.4.3 AN/AQA-7(V) Tests - The equipment shall generate and provide test signals to verify the complete operation of the active modes of the AN/AQA-7(V) systems. (See 3.4.8.)

3.3.7.4.4 Computer Test - The equipment shall provide for the following methods of fault detections and fault isolation by aircraft computer.

(1) Integration test
(2) Systems GO/NO-GO test (SYGNOG)
(3) Diagnostic tests

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3.3.7.4.4.1 Integration Tests - Integration tests performed by the computer shall be used to test the following:

(1) Communication paths between the equipment and the computer

(2) Functional capabilities of the equipment when under computer control

(3) Basic timing of the equipment

3.3.7.4.4.2 Systems GO/NO-GO Test (SYGNOG) - SYGNOG tests performed by the computer shall be primarily a fault detection test to determine functional availability and capability of the equipment.

3.3.7.4.4.3 Diagnostic Tests - Diagnostic tests performed by the computer shall provide fault location to the WRU of the equipment.

3.3.7.5 Peculiar Support Equipment - The equipment shall not require PSE (Peculiar Support Equipment) to perform organizational level maintenance actions. General purpose support equipment, i.e., meter, oscilloscopes, will be allowed in the performance of organizational maintenance actions.

3.3.7.6 Packaging Requirements - The equipment shall be packaged in modular form so that all subassemblies shall be of the plug-in module type and are easily removable without special tools for maintenance. If a special tool is required for removal, it shall be provided and attached to the equipment.

3.3.7.6.1 Module Arrangement - Module arrangement will be such that access to each module does not require the removal of adjacent modules or parts other than access panels.

3.3.7.6.2 Keying - Modules and connectors shall be keyed to prevent the insertion of a connector or module into an improper location within or on the equipment at the organizational level.

3.3.7.6.3 Access - It shall be possible to perform 90% of the organizational maintenance actions by access to one side of the equipment.

3.3.7.6.4 Access Panels - Access panels shall be retained by quick disconnect fasteners.

3.3.7.6.5 Loose Hardware - Devices requiring removal for inspection and/or servicing shall be retained by quick disconnect fasteners.

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Non-captive hardware (screws, nuts, washers) are strictly forbidden in those areas where organizational maintenance may be required.

3.3.7.6.6 Aircraft Connectors - Aircraft cables shall mount directly to the equipment, using quick disconnect type connectors except the RF antenna connector on unit 3. (See figure 3.)

3.3.7.7 Protective Devices - Circuit breakers shall be used in lieu of fuses whenever possible. Fuses, if used, shall be of types and ratings readily available from Navy Supply. Fuse holders and/or circuit breakers shall be accessible from the exterior of the equipment, plainly marked as to rating and size, and shall be of the fault indicating type and/or have access for quick voltage checks.

3.3.7.8 Test Points

3.3.7.8.1 Test Point Requirements - Test points shall be provided to facilitate rapid localization of malfunctions and to allow efficient maintenance, repair, calibration, alignment, checkout, and monitoring of the system. Test point definition found in MIL-STD-415 shall apply.

3.3.7.8.2 Power Supply Test Points - Exposed test points (see Definitions, MIL-STD-415) shall be provided on the equipment to measure significant voltages. Test points shall terminate in pin jacks, sockets, or connectors suitable for use with oscilloscope and multipurpose meter probes. Multipin connectors will be acceptable provided an adapter (breakout box) suitable for use with oscilloscope and meter probes is included as an integral part of the system.

3.3.7.8.3 Fault Isolation Test Points - Exposed test points shall be provided for use at the organizational level of maintenance, in isolating faults beyond the capability of the BITE, or when other provisions have not been made for fault isolation to the quick replaceable assembly (module) level. In addition, for that equipment incorporating BITE, test points shall be provided to repair the BITE in the event of failure. Test points shall be accessible from the front of the equipment, and shall terminate in pin jacks, sockets, or connectors suitable for use with oscilloscope and meter probes. Where practical each test point shall be marked in a durable and readily visible fashion showing waveforms, voltages, and other signal characteristics normally applicable to that pin.

3.3.7.8.4 Shop Repair Test Points - Test points shall be provided on the equipment for use at the intermediate level of maintenance in isolating faults which cannot be isolated at the organizational level or which cannot be isolated to a quick replaceable assembly (e.g., module). The test points shall provide means for isolating faults, for calibrating and aligning the equipment, and for introducing any other necessary functional inputs or dummy loads necessary for bench test.

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3.3.7.9 WRU (Weapons Replaceable Unit - For the purpose of fault isolation, the WRU's are:

(1) Control, reference signed generator
(2) Reference signal generator
(3) Each plug-in module of (2) above
(4) Radio transmitter

3.3.7.10 Changes to AR-10 -

(a) Change all references to Weapons Replaceable Assembly to Weapons Replaceable Unit (defined in USAS Y32.16-1965).

(b) Change all references to Shop Replaceable Assembly to Replaceable Assembly.

3.3.7.11 Maintainability Verification - Test requirements shall be established under Quality Assurance provisions of this specification.

3.3.8 Nomenclature and Nameplates - Nomenclature and serial number assignment, nameplate approval and equipment identification shall be in accordance with Specification MIL-N-18307.

3.3.9 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room Ambient (25° C ± 5° C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input Power	(a) AC power (three phase) 208 v, ± 1 volt
	(b) AC power (single phase) 28 v, +0 -10 volts
	(c) DC power, 4 vdc, +0.2 -0.8 volt
	(d) DC power, +11 v, ±1 volt
	(e) DC power, +21 v, ±1 volt
	(f) DC power, -21 v, ±1 volt

3.3.10 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for class 1AX equipment, except as modified herein.

3.3.10.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of curve IV of Specification MIL-E-5400.

3.3.10.2 Temperature - The equipment operating extreme temperatures shall be -30° to +55° C including the intermittent operating tests.

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3.3.10.3 Altitude - The equipment operating maximum altitude shall be 30,000 feet.

3.3.11 Secondary Input Power Requirements - The power required shall not exceed the specified amounts. The secondary power will be supplied by the AN/AQA-7(V) system.

(a) AC power (three phase) 208 v, category B, 100 va max

(b) AC power (single phase) 28 v nominal (18-28 v), 0.6 ampere max

(c) DC power, +5 vdc nominal (4.2 to 5.2 v), 4 amperes max

(d) DC power, +11 vdc, 8 amperes max

(e) DC power, +21 vdc, 2 amperes max

(f) DC power, -21 vdc, 2 amperes max

3.3.11.1 Degraded Performance - Degraded performance will be permitted when the Primary Input Power is subjected to voltage transients from 0 to 0.5 second duration as shown in figure 2 of MIL-STD-704 for Normal Electric System Operation. Operation shall return to normal when the 0 to 0.5 second transient expires. Protection against short pulses (50 microseconds) will be provided by unit 11 of the AN/AQA-7(V).

3.3.11.2 Overload Protection - Overload protection will be provided by unit 11 of the AN/AQA-7(V). All parts and circuits of the equipment which are likely to carry on overload due to any failures or poor adjustments shall be proportioned to withstand such overload without permanent damage to the equipment or shall have suitable protective devices. The use of fuses and other protective devices shall be held to a minimum.

3.3.11.3 Undervoltage Protection - The equipment shall not be damaged by voltages below the minimum specified herein and shall automatically resume normal operation when the voltage returns within limits.

3.3.11.3.1 Voltage Regulators - Voltage regulators shall be provided to regulate the unregulated dc power specified in paragraph 3.3.11 to the voltage levels required for circuit operation.

3.3.11.4 Lighting Power - Input power for lighting shall not require more than 2 amps at 5.0 volts.

3.3.11.5 Warmup Time - The time required for the equipment to warm up prior to operation shall be kept to a minimum and shall not exceed 1 minute under standard conditions and 2 minutes at extreme service conditions.

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3.3.11.6 Cooling - The cooling of the equipment shall comply with the requirements of Specification MIL-E-5400. The thermal design tests shall be conducted as required by Specification MIL-T-23103.

3.4 Performance - Unless otherwise specified, values set forth to establish the requirements of satisfactory performance apply to performance under both standard and extreme service and input power conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified. The specified performance values shall be attained when power, control, and signals are provided by the AN/AQA-7(V).

3.4.1 Operation - The equipment shall generate the necessary signals to control the operation of floating sonic devices and permit processing of their returned signals. Operation of the equipment shall be either computer controlled or manually controlled at the discretion of the operator. The signals and control functions shall be classified as follows:

Function	Applicable Paragraph
(a) Radio Frequency Generation	3.4.2
(b) Modulation Tone Generation	3.4.3
(1) Address Generation	
(2) Command or Sonic Generation	
(3) UHP-2 Modulation	
(c) Received Signal Processing	3.4.4
(d) Manual Control	3.4.5
(e) Computer Control	3.4.6
(f) External Mode	3.4.9

3.4.1.2 Impedance and Voltage Levels - Digital and audio signal impedance levels offered to intra and interconnected units shall be as in table I.

TABLE I

IMPEDANCE AND VOLTAGE LEVELS

Signal	AS Load			AS Source		
	Volt	DC	AC	Volt	DC	AC
Digital (Logic 1	0 ± 0.5	>2KΩ	130Ω	0 ± 0.5	≤200Ω	
Digital (Logic 0)	4 ± 1			4 ± 1		
All others			10KΩ		≤1KΩ	
Audio Transmitter		100Ω ± 5Ω			≤1KΩ	

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3.4.2 Radio Frequency Generation

3.4.2.1 Operation - The equipment shall generate modulated rf signals suitable for transmission with the following characteristics.

3.4.2.2 Frequency Characteristics - The equipment shall provide a carrier frequency of 291.4 MHz ±0.005%.

3.4.2.3 Modulation Characteristics - The rf carrier shall be capable of being amplitude modulated with signals from 1 kHz to 50 kHz.

3.4.2.3.1 Percentage of Modulation - The rf carrier shall be amplitude modulated 90 ±5% when the input voltage from the unit 2 is adjusted to 0.5 ±0.025 VRMS terminated in an impedance of 100 ohms or less.

Percent modulation is determined from measurement of maximum and minimum voltages displayed on an oscilloscope and from the following formula: $\frac{Emax - Emin}{Emax + Emin} \times 100$ - Percentage Modulation
Under extreme service conditions, modulation shall not be less than 80 percent.

3.4.2.3.2 Distortion and Spurious Signals - The modulated rf carrier shall not contain distortions or spurious signals in excess of the following limits.

(a) Harmonic Distortion	5% max in frequency range of 5,000 to 10,000 Hz, 10% max for frequencies from 10,000 to 50,000 Hz.
(b) Intermodulation Distortion	The level of the sum and difference frequencies of two equal livel discrete modulating signals shall be 25 db below either modulating signal.
(c) Noise Modulation	-40 db below normal modulation levels.
(d) Frequency Modulation	±100 Hz peak with normal modulation levels.

3.4.2.4 Transmitted Power Output - The equipment shall deliver not less than 12 watts of unmodulated carrier into a 50 ohm resistive load and 10 watts

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of unmodulated carrier into a 50 ohm resistive load under extreme service condition.

3.4.2.4.1 Output Load Mismatch - The equipment shall deliver at least 10 watts forward unmodulated carrier power when terminated by a load having a VSWR greater than 1.0 to 1.0 but equal to or less than 1.5 to 1. The equipment shall not be damaged by transmission into any load including a short or open antenna circuit.

3.4.2.4.2 Transmitter Input Termination - The audio input circuit of the radio transmitter shall have a characteristic impedance such that the unit 2 output remains constant in voltage.

3.4.2.5 Phase Linearity - The phase response shall be linear with frequency within ±15 degrees throughout the sonar spectrum as specified in paragraph 3.4.4.1.4 for the 3 db point. In addition, the phase versus frequency slope shall not exceed 3.5 degrees per Hz.

3.4.3 Modulation Tone Generation -

3.4.3.1 Operation - The equipment shall modulate the carrier with two sequential tones. The first part shall contain the address tones and the second part shall contain either a command tone or sonic tone. The characteristics of the modulating tone to the transmitter shall be as specified in 3.4.3.2 and 3.4.3.3.

3.4.3.2 Modulating Tones -

3.4.3.2.1 Duration - The duration of the tones shall be as specified in table II.

TABLE II

MODULATING TONE TIMING (IN MILLISECONDS)

Mode	Address	Guard Interval	Sonic/Command
Active* (CW-1.0)	250 ±25	2.5 ±0.5	1,000 ±10
Active* (CW-0.5)	250 ±25	2.5 ±0.5	500 ±5
Active* (CW-0.1)	250 ±25	2.5 ±0.5	100 ±2
Active* (FM)	250 ±25	2.5 ±0.5	1,000 ±10
Active* (Command)	250 ±25	2.5 ±0.5	1,000 ±10
Passive (Command)	250 ±25	2.5 ±0.5	1,000 ±10

*Active AN/SSQ-50 or AN/SSQ-62

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3.4.3.2.2 Modulating Tones Repetition Rates (Computer Control - Unit 3 shall be capable of continuous duty for 2.0 minutes followed by 4.0 minutes of 50% duty cycle; however, during any 10 minute interval the duty cycle shall not exceed 50%.

3.4.3.2.3 Modulating Tones, Repetition Rates (Manual Control) - When under manual control the equipment shall be capable of operating continuously at a duty cycle of one transmission followed by 2 seconds standby. When transmitting four sequential commands the cycle times shall be:

(a) 16 ±0.2 seconds
(b) 32 ±0.4 seconds

3.4.3.3 Frequency of Modulating Tones - The equipment shall modulate the carrier for the address duration with two simultaneous tones and modulate the carrier with a single tone for the command or sonic duration as specified herein.

3.4.3.3.1 Tone Frequency, Address (Computer) - When under computer control the equipment shall generate and modulate the carrier with two tones as prescribed by a single computer instruction word. These addresses shall be the same as in the manual mode.

3.4.3.3.2 Tone Frequency Address (Manual) - When under manual control the equipment shall generate and modulate the carrier for the address duration (3.4.3.2) table II with address tones specified in (3.4.3.3.3) table III.

3.4.3.3.3 Address Frequency Stability - The frequency of the tones generated in either mode shall not vary more than ±0.05 percent from that specified.

3.4.3.3.4 Address Tone Balance - Each tone shall contribute 0.250 ±0.025 VRMS to the combined address tone.

3.4.3.3.5 Command Tones - Command tones shall be generated with the COMMAND CONTROL set for active or passive and the the PING RATE CONTROL set for manual. The single tone frequency shall be as follows:

Command	Tone Frequency
VHF on/off	27 kHz ±13.5 Hz
Scuttle	21 kHz ±10.5 Hz
Depth Select	33 kHz ±16.5 Hz

3.4.3.3.6 Sonic Tones - Sonic tones shall be generated in all modes in accordance with table IV.

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TABLE IIIA

ACTIVE ADDRESS TONES

Sonobuoy VHF Channel	Address Tones in Hz f_1	f_2
1	12,700	13,400
2	12,700	14,200
3	12,700	15,300
4	12,700	16,100
5	12,700	17,100
6	12,700	18,000
7	13,400	14,200
8	13,400	15,300
9	13,400	16,100
10	13,400	17,100
11	13,400	18,000
12	14,200	15,300
13	14,200	16,100
14	14,200	17,100
15	14,200	18,000
16	15,300	16,100
17	11,800	12,700
18	11,800	13,400
19	11,800	14,200
20	11,800	15,300
21	11,800	16,100
22	11,800	17,100
23	11,800	18,000
24	19,000	11,800
25	19,000	12,700
26	19,000	13,400
27	19,000	14,200
28	19,000	15,300
29	19,000	16,100
30	19,000	17,100
31	19,000	18,000

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TABLE IIIB

PASSIVE ADDRESS TONES

Sonobuoy VHF Channel	Address Tones in Hz	
	f_1	f_2
1	26,300	27,100
2	26,300	29,100
3	26,300	30,500
4	26,300	31,900
5	26,300	33,300
6	26,300	34,400
7	26,300	35,800
8	26,300	37,200
9	27,100	29,100
10	27,100	30,500
11	27,100	31,900
12	27,100	33,300
13	27,100	34,400
14	27,100	35,800
15	27,100	37,200
16	29,100	30,500
17	29,100	31,900
18	29,000	33,300
19	29,100	34,400
20	29,100	35,800
21	29,100	37,200
22	30,500	31,900
23	30,500	33,300
24	30,500	34,400
25	30,500	35,800
26	30,500	37,200
27	31,900	33,300
28	31,900	34,400
29	31,900	35,800
30	31,900	37,200
31	33,300	34,400

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TABLE IV

SONIC TONE TABLE

Sonar Channel	Sonobuoy VHF Channels	Center Frequency
A	1, 5, 9, 13, 17, 21, 25, 29	6.5 kHz
B	2, 6, 10, 14, 18, 22, 26, 30	7.5 kHz
C	3, 7, 11, 15, 19, 23, 27, 31	8.5 kHz
D	4, 8, 12, 16, 20, 24, 28	9.5 kHz

3.4.3.3.7 CW Frequency Stability - The frequency accuracy shall be plus or minus 2 Hz from the sonar channel frequency as given in table IV and with a stability of plus or minus 0.3 Hz during any 20 second period.

3.4.3.3.8 FM Linearity and Stability - There shall be a total linear sweep of 400 Hz symmetrical about the sonar channel center (table IV) frequency. The direction of the frequency sweep shall be selectable for either an increasing or decreasing frequency as a function of time. The frequency deviation from a linear function shall be less than or equal to ±1.0 Hz during a single sweep. The frequency at the start of any sweep shall be within 2.0 Hz of the assigned value.

3.4.3.3.9 Pulse Rise Time and Fall Time - The rise and fall times of the address, sonic and command tones shall not exceed 5 milliseconds when measured between the 5 and 90 percent amplitude points.

3.4.3.3.10 Sonic/Command Tones Amplitude - The amplitude of the sonic/command tones shall be 0.5 ±0.025 VRMS.

3.4.3.3.11 Modulation Tones Pulse Droop - The modulation tone pulse shall be flat within the pulse length as specified in table II of paragraph 3.4.3.2.1 and of an amplitude as specified in paragraphs 3.4.3.3.4 and 3.4.3.3.10. The pulse droop shall not exceed 1 db measured 10 milliseconds after the start of the tone pulse to 10 milliseconds before the end of the tone pulse.

3.4.3.4 UHF-2 Radio Transmitter, Modulation - The equipment shall provide signals to modulate the UHF-2 radio transmitter.

(a) Modulation coding shall be as specified in 3.4.3.

(b) The amplitude of the tones (either single or double) shall be 1.414 ±0.07 volts peak to peak.

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(c) The rise and fall time of the tones shall not exceed 10 milliseconds when measured between the 10 and 90 percent amplitude points.

(d) A digital keying signal shall be provided 2.5 milliseconds before the start of address tone and end with the termination of the sonic or command transmission. The impedance and voltage levels shall be as defined in paragraph 3.4.1.2 with rise and fall times measured between 10% and 90% amplitude points of less than 500 usec.

(e) A remote UHF on enable and remote UHF on enable return shall be supplied for enabling outputs as defined in (a) through (d) above. When the two signal lines are shorted, the functions (a) through (d) shall be enabled. Open lines shall disable the outputs.

3.4.4 Received Signal Processing - The equipment shall process received signals and provide signals for further processing as specified herein. The processing shall be classified as follows:

(a) AN/SSQ-47
(b) AN/SSQ-50
(c) AN/SSQ-62

3.4.4.1 AN/SSQ-50 Processing -

3.4.4.1.1 AN/SSQ-50 (CW) - When operating in the CW mode the equipment shall heterodyne the received signal to 800 Hz +2.0 Hz nominal center frequency ± the target doppler.

3.4.4.1.1.1 CW Notch Filter - When operating in the CW mode the input signal shall at the operator's selection on unit 1 pass through the notch filter located in unit 2 and after the sonar signal switching matrix. Thus it shall be possible to display a sonobuoy simultaneously with and without the notch filter using the upper and lower trace of a sonar sensor station of the AN/AQA-7(V).

3.4.4.1.1.2 Center Frequency - The notch filter maximum attenuation shall occur at a center of 800 Hz ±3 Hz and shall be greater than 50 db.

3.4.4.1.1.3 Attenuation Characteristics - The notch filter shall have fixed (non-adaptive) characteristics and the plot of attenuation as a function of frequency shall fall within the best straight line approximation of the area defined by table V.

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TABLE V

NOTCH FILTER CHARACTERISTICS

Attenuation DB	Frequency Limits	
	Maximum	Minimum
3	760	725
6	770	745
10	780	763
20	790	782
30	795	789
40	798	792
40	808	802
30	811	805
20	818	810
10	840	823
6	860	835
3	890	850

3.4.4.1.2 AN/SSQ-50 (FM) - When operating in the FM mode the equipment shall heterodyne the input to 800 ±2.0 Hz nominal center frequency ± the target doppler.

3.4.4.1.3 Dynamic Range - The dynamic range of the frequency converter shall be linear within ±1.0 db and have unity gain ±1 db over an input signal range of 10 mv RMS to 2 VRMS. The frequency converter shall be capable of passing a maximum input signal of 2.5 VRMS.

3.4.4.1.4 Frequency Response - The frequency response shall be determined by the maximum deviation of the FM sonic pulse from the sonar channel center frequency plus the doppler frequency shift corresponding to a maximum radial target velocity of 35 knots. The output -3 db, -10 db, and -20 db points versus input frequency are shown in table VI.

TABLE VI

TRANSFER CHARACTERISTICS

Sonar Channel	f_1 (Hz) -20 db Point	f_2 (Hz) -10 db Point	f_3 (Hz) -3 db Point	f_4 (Hz) Center Frequency	f_5 (Hz) -3 db Point	f_6 (Hz) -10 db Point	f_7 (Hz) -20 db Point
A	5500	5900	6148	6500	6861	7135	7600
B	6400	6850	7123	7500	7886	8175	8700
C	7350	7825	8099	8500	8910	9225	9800
D	8200	8775	9075	9500	9934	10,250	10,850

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3.4.4.1.5 Phase Response - The phase response of each input frequency converter, in the FM mode, shall be linear within 15 degrees for the frequency bands specified below.

Sonar Channel	f_3 + 5% BW Lower Frequency	f_5 - 5% BW Upper Frequency
A	6183 Hz	6825 Hz
B	7160 Hz	6848 Hz
C	8139 Hz	8870 Hz
D	9117 Hz	9891 Hz

3.4.4.2 AN/SSQ-47 Processing - When operating in the AN/SSQ-47 mode the equipment shall separate the trigger signal from the sonar signal. A trigger gate shall be generated when the trigger signal level is 0.5 VRMS or more and the trigger frequency lies between 26 and 38 kHz. The maximum leading and trailing edge delays of the trigger gate shall not exceed 5 milliseconds.

3.4.4.2.1 Dynamic Range of Input Circuits - The dynamic range of the input circuit for processing the sonar signal shall be linear within ±1 db and have unity gain ±2 db over an input signal range of 10 millivolts RMS to 2 volts RMS. The input circuit shall be capable of passing a maximum input signal of 2.5 VRMS.

3.4.4.3 AN/SSQ-62 Processing - When operating in the DICASS omni select mode the equipment shall demultiplex the omni-channel from four AN/SSQ-62 sonobuoys and supply these signals to the AN/AQA-7(V) for processing. In addition, the equipment shall supply mode controls to the AN/AQA-7(V). The demultiplex omni-channel data shall be supplied and switchable to the two AN/AQA-7(V) systems.

3.4.4.3.1 Dynamic Range of Input Circuits - The dynamic range of the input circuit for processing the composite signal shall be linear within ±1 db and have unity gain ±2 db over an input signal range of 10 millivolts to 2 volts RMS. The input circuit shall be capable of passing a maximum input signal of 2.5 volts RMS.

3.4.4.4 Tape Replay Trigger Tone - A trigger tone shall be generated to coincidence with the sonic gate for both the active and the AN/SSQ-47 mode of operation.

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3.4.4.4.1 Trigger Tone to Tape Recorder - A 3.2 kHz ±100 Hz trigger tone shall be generated and gated into the tape recorder input line of the sonar channel being used. The trigger tone shall be coincident with the sonic gate of its respective channel. The tape recorder input signal shall contain both the trigger tone and the sonic signal. The trigger tone shall have characteristics as follows:

(a) Leading and trailing edge delay (with respect to sonic gate); 2 milliseconds

(b) Amplitude; from 1.0 VRMS to 2 VRMS

(c) Harmonic content; ≤15%

3.4.4.4.2 Trigger Tone From Tape Recorder - A replay mode shall be provided enabling replay of four channels of recorded sonic information. Provision shall be made to generate a sonic gate when the recorded signal contains a 3.2 kHz ±100 Hz trigger tone at a minimum amplitude of 0.5 VRMS. The maximum leading and trailing edge delays of the sonic gate shall not exceed 5 milliseconds. The replay mode shall be remotely enabled only. Unit 1 shall provide mode enables during tape replay.

3.4.4.5 Controls to Aircraft Data Processor and Demultiplexer - The equipment shall provide the required mode and control signals to the aircraft data processor and demultiplexer for the functions listed in paragraph 3.4.5. Separate CASS, DICASS, CW and FM mode select will be provided for each SSS (Sonar Sensor Station). When both SSS' are displaying the same sonobuoy, the SSS II CW and FM mode select will automatically track SSSI CW and FM mode select.

3.4.5 Manual Controls - The unit 1 shall provide manual selection of the following functions:

(a) Mode Select

AN/SSQ-47 - Routes the returned signal from sonobuoy receivers without additional processing to the AN/AQA-7(V) processor and display except as noted in paragraph 3.4.4.2.

OMNI/DICASS - There shall be one OMNI/DICASS selection for each sensor station. When OMNI is selected, the returned sonar signals (via the sonobuoy junction box) shall be heterodyned in the AN/ASA-76 prior to being routed to the AN/AQA-7(V) via the sonobuoy junction box, unless a sonar channel has been placed in the OMNI/DICASS mode as defined in (j). When DICASS is selected for one or both sensor stations, signals shall be supplied to

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program the AN/AQA-9 as to sensor station(s) selected for DICASS, sonic mode and event start. When either OMNI or DICASS is selected the equipment shall supply address and sonic tones to the active sonobuoys as specified herein.

Computer - Transfer control of the equipment from the sonar operator to the aircraft computer.

(b) CW Mode Select

0.1 second - Provides 100 millisecond CW sonic pulse as described in tables II and IV and paragraphs 3.4.3.3.6 and 3.4.3.3.7.

0.5 second - Provides 500 millisecond CW sonic pulse as described in tables II and IV and paragraphs 3.4.3.3.6 and 3.4.3.3.7.

1.0 second - Provides 1000 millisecond CW sonic pulse as described in tables II and IV and paragraphs 3.4.3.3.6 and 3.4.3.3.7.

(c) FM Mode Select

Up - Provides a FM sonic pulse increasing in frequency as a function of time as described in paragraph 3.4.3.3.8.

Down - Provides a FM sonic pulse decreasing in frequency as a function of time as described in paragraph 3.4.3.3.8.

(d) Command Select - Allows the sonar operator to select one VHF channel and permits the equipment to address active or passive sonobuoys and supply any one of the four command functions described below. The initiation of any of these functions shall not inhibit the equipment from the addressing and the sonic commanding of any sonobuoy while in one of the auto ping rates. The execution of any one of these functions will be performed on a non-interference basis if the equipment is in one of the auto ping rates.

VHF on/off - Provides a command tone as described in paragraph 3.4.3.3.5 to enable or disable the VHF transmitter of the sonobuoy.

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Scuttle - Provides a command tone as described in paragraph 3.4.3.3.5 to scuttle the selected sonobuoy.

Depth - Provides a command tone as described in paragraph 3.4.3.3.5.

External - When selected transmits address tones and then allows external equipment to modulate the carrier of unit 3 for three seconds.

(e) BIT - When selected energizes the BUILT-IN-TEST.

(f) Fault - When BIT has been selected and a fault is detected this indicator will be illuminated.

(g) Ping Rate Control

Manual - When selected permits the sonar operator to control the ping rate of the sonic pulse as described in paragraph 3.4.3.2.3.

16 ±0.2 Second Automatic - When selected the equipment automatically sequences through the four sonar channels and in the sonic mode selected; transmitting the address and sonic tones for only those sonobuoys addressed. It will take 16 ±0.2 seconds to accomplish the cycle. The order of addressing and sonic commanding will be as follows: (1) sonobuoy selected for SSS No. 1 upper, (2) sonobuoy selected for SSS No. 2 upper, (3) sonobuoy selected for SSS No. 1 lower, (4) sonobuoy selected for SSS No. 2 lower. If a sonobuoy is selected for displaying by more than one SSS it will be displayed simultaneously by the traces which have selected this sonobuoy.

32 ±0.4 seconds - When selected the equipment automatically sequences through the four sonar channels and in the sonic mode selected; transmitting the address and sonic tones for only those sonobuoys addressed. It will take 32 ±0.4 seconds to accomplish the cycle.

(h) Buoy Select - Permits the sonar operator to select one sonobuoy VHF channel of each of the four sonar channels, (4 of 31).

(i) Manual Ping - There shall be four manual controls. One shall be associated with each of the display traces, i.e., sensor station

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1 upper, sensor station 1 lower, sensor station 2 upper, and sensor station 2 lower. These controls shall be operable only in the PING RATE manual mode. When any one of these controls are manually initiated, information concerning all display traces programmed with that sonar channel shall be programmed into unit 2 for supplying mode controls, ping gates and channel switching to the AN/AQA-7(V) systems. In addition the program address (associated with that sonar channel) and sonic tone shall be transmitted to the active sonobuoy as specified herein.

(j) Omni-DICASS Select - Operator may select any of the four sonar channels to be demultiplexed and processed in the omni search mode when a DICASS sonobuoy is being interrogated.

(k) Sonar Sensor Station 2 - Display Select Upper - Permits the sonar operator to route one of the four sonar channels returned signals to the upper trace of the display of the AN/AQA-7(V). The sonar signal supplied to the tape recorder via the aircraft sonobuoy junction box shall not pass through the switching matrix.

Display Select Lower - Permits the sonar operator to route one of the four sonar channel returned signals to the lower trace of the display of the AN/AQA-7(V). When DICASS is selected, it tracks the display select upper.

(l) Sonar Sensor Station 1 - Display Select Upper and Lower has the same function as those for Sensor Station 2.

(m) Notch Filter - When selected the returned sonar signal to the AN/AQA-7(V) passes through the notch filter described in paragraph 3.4.4.1.1.1. The sonar signal supplied to the tape recorder via the aircraft sonobuoy junction box shall not pass through the notch filter. Four notch filter controls will be provided, one per processing channel.

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3.4.6 Computer Control - When the computer mode is selected on unit 1, the equipment shall receive instruction from the aircraft computer (CP-901) via the DIU (Digital Interface Unit) of the AN/AQA-7(V). Disconnection of the control panel will automatically place the AN/ASA-76 in the computer mode of operation. In the computer mode, the equipment shall be capable of performing all the functions which are programmable from unit 1 except for the ping rate which will be dependent upon the command rate from the computer as specified in paragraph 3.4.3.2.2.

3.4.6.1 Digital Communications

3.4.6.1.1 Digital Command and Data Transfer - Digital commands and data transfers between the DIU and the Computer are by parallel transmission of data and control as described in the Program Technical Memorandum for Sonar Computer-Recorder AN/AQA-7(V) (U) (P-3C aircraft) Volume I of 15 June 1969. Digital command and data transfers between the DIU and the equipment are serial transmission of data and control as described in the Program Technical Memorandum and by this specification.

3.4.6.1.2 Input/Output Word Response - The equipment shall generate and respond to the computer input/output words function described in this specification.

3.4.6.1.3 Computer Word Formats and Codes - The computer word formats shall be supplied and are subject to the approval of the procuring activity.

3.4.6.1.4 Coding Formats - Standard encoding formats should be used. Such codes should be simple in structure and easily related to the function to be performed. Complex or irregular codes should be avoided. Wherever possible, use of all zeros codes for functional commands should be avoided. All zeros codes should be used where no action or no change conditions are desired.

3.4.6.1.5 Illegal Codes - Codes not having specific functions regarding the equipment performance shall be considered illegal codes and the equipment shall not respond.

3.4.6.1.6 Coding Simplicity - Computer words should be formatted such that codes for the functions to be performed are arranged in octal groups, and can be generated by a simple algorithm, i.e., simple sequencies and progressions.

3.4.6.1.7 Reserved Bits - Bit 0 is a reserved bit to identify the input/output word as a AN/ASA-76 word (bit 0 - logical 1) or as an AN/AQA-7(V) word (bit 0 - logical 0). There are no other reserved bits for the computer input word, however, bits 27, 28 and 29 of the computer output word are reserved

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for AN/ASA-76, and in the AN/ASA-76 mode must contain a Logical 1 in one or more of these bits if a return word from AN/ASA-76 is expected.

3.4.6.2 Computer Output Word

3.4.6.2.1 Output Word Types - The equipment shall respond to the following three types of computer output words from the aircraft computer.

(1) Command Word
(2) Return a Status Word
(3) Return a BIT Word

3.4.6.2.2 Computer Output Word Identification - The computer output word shall contain the necessary codes and words to identify the output word type.

3.4.6.2.3 Command Word - The Command Word shall control the equipment when equipment is in the COMPUTER MODE.

3.4.6.2.3.1 Command Word Application - The Command Word is a Computer output word and shall be used when the computer requires a change in the AN/ASA-76 operating modes, display modes and/or emit a command to the AN/SSQ-50, An/SSQ-62 sonobuoys or passive command configured sonobuoys via UHF command link.

3.4.6.2.3.2 Response to Command Word in Manual Mode - The equipment shall ignore instructions contained in the Command Word when the equipment is in one of the manual modes except that upon receipt of the Command Word, the equipment shall respond with a Control Panel Status Input Word. Manual modes are herein defined as one of the following:

(1) MANUAL
(2) AUTO 16
(3) AUTO 32
(4) Q-47

3.4.6.2.3.3 Command Word Structure - The Command Word shall be a one word transfer. The Command Word shall contain all the necessary fields and codes to duplicate the switch functions of Unit 1 with the following exceptions:

(1) The Command Word cannot switch to COMPUTER mode
(2) The Command Word cannot switch to MANUAL mode
(3) The Command Word cannot operate the auto 16 or auto 32 function

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3.4.6.2.3.4 Command Word Capability - The Command Word shall be capable of performing the following:

3.4.6.2.3.4.1 RF Channel Selection; any of 31 RF channels

3.4.6.2.3.4.2 Sonobuoy Type Selection; passive or active

3.4.6.2.3.4.3 Selection of Command to be executed;

(1) VHF carrier on-off
(2) hydrophone depth
(3) scuttle
(4) external modulation
(5) sonic ping

3.4.6.2.3.4.4 Selection of Sonic Ping/Processor Modes

3.4.6.2.3.4.4.1 Ping Modes

(1) CW 0.1 Sec
(2) CW 0.5 Sec
(3) CW 1.0 Sec
(4) FM Up
(5) FM Down
(6) No Ping

3.4.6.2.3.4.4.2 Processor Modes

(1) CW
(2) FM
(3) DICASS
(4) Omni DICASS
(5) SSQ-47
(6) No Mode Change

3.4.6.2.3.4.4.3 Ping Mode Select - For each Command Word there shall be one Ping or Command Function Mode selected.

3.4.6.2.3.4.4.4 Pre-Processor Mode Select - For each Command Word there shall be a Processor Mode selected. When the SSQ-47 mode is selected, both sensor stations shall operate in that mode.

3.4.6.2.3.4.4.5 Pre-Processor Pairs - Sensor Station 1 shall provide for pre-processing Sonar Channel A and B. Sensor Station 2 shall provide for pre-processing Sonar Channel C and D.

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3.4.6.2.3.4.5 Selection of Sonar Channel/Display Channel Assignment - The Command Word shall be capable of assigning any Sonar Channel (A, B, C, or D) to any of the active display areas associated with a two set AN/AQA-7(V) system. In no case shall a Command Word assign more than 1 Sonar Channel to a particular display area at a time. When DICASS Mode is selected, each Sensor Station Display shall display the following format for the selected Sonar Channel:

(1) Upper trace - Range vs. Doppler
(2) Lower trace - Range vs. Bearing

3.4.6.2.3.4.6 Selection of Notch Filter - The Command Word shall be capable of inserting or removing the notch filter from each of the 4 displays. The notch filter selection shall be independent for each display channel.

3.4.6.2.3.4.7 No RF Emission - The Command Word shall have the capability of causing Sonic Ping/Processor Mode and Display Mode changes to the equipment without causing RF emission. An RF EMISSION INHIBIT function shall be in effect when a NO CHANNEL code is contained in the RF channel field of the Command Word.

3.4.6.2.3.4.8 No Channel Code - Each Command Word must contain a RF channel address or the NO CHANNEL code.

3.4.6.2.3.4.9 No Change Code - The Command Word shall have the capability of NO CHANGE Codes for Sonic Ping/Processor and Display modes. Additionally, each Command Word shall have the capability of independently applying the NO CHANGE code to either sensor station or any display.

3.4.6.2.3.4.10 Buoy Type Code - Each Command Word that has a RF channel address must contain a buoy type code (passive, active).

3.4.6.2.3.4.11 Instantaneous Abort of Transmission - The Command Word shall have the capability of INSTANTANEOUS ABORT OF TRANSMISSION. Upon receipt of this code, the RF emission shall be immediately terminated. This command shall cause no change to Display Modes or Sonic Ping/Processor Modes.

3.4.6.2.3.4.12 Command Initiation - Any function to be performed by the equipment when in Computer Mode shall be initiated within a maximum time of 0.25 msec from receipt of the Command Word by the DIU.

3.4.6.2.3.4.13 Timing Constraints - There shall be no timing constraints placed on the Computer generated command or Return Status Word(s) other than the time required for Command and Return Status Word transfers.

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3.4.6.2.4 Return Status Word -

3.4.6.2.4.1 Return Status Word Application - A Return Status Word is a Computer output word which shall be used to interrogate the equipment as to its operating modes and functions. Receipt of a Return Status Word by the equipment shall cause the equipment to return the following status to the Computer via the DIU.

(1) status of the last function programmed into Unit 2 by Unit 1 when the AN/ASA-76 is in one of the manual modes.

(2) when the equipment is in the COMPUTER mode, a status of the last Command Word received by the equipment; this requirement applies only when the Return Status Word is preceded by a Command Word.

3.4.6.2.4.2 No Emission of RF - The computer output of a Return Status Word shall not cause emission of RF from CASS or change of a Sonic Ping/Processor or Display Mode.

3.4.6.2.4.3 Return Status Word Transfer Rate - The transfer of Return Status Word from Computer to the equipment shall not exceed 0.25 msec from the time the DIU receives the Return Status Word.

3.4.6.2.5 Return BIT Word

3.4.6.2.5.1 Return BIT Word Application - The Return BIT Word is a Computer output test word which shall be used to determine availability and performance of the equipment functions. The Return BIT Word shall cause specific tests to be performed on the equipment by exercise of BITE (Built-In Test Equipment).

3.4.6.2.5.2 Return BIT Word/BIT Word Sequence - For each Return BIT Word there shall be an associated BIT Word response to the Computer. The BIT Word shall contain the results of the specific tests performed.

3.4.6.2.5.3 Return BIT Word Identification - Each Return BIT Word shall contain the specific mode or function under test.

3.4.6.2.5.4 BIT Emission - Emission of RF from the command transmitter is permissible when the equipment is being operated under control of the Return BIT Word.

3.4.6.2.5.5 Return BIT Word/BIT Word Transfer Rate - 3.0 sec is the allowable time for a Return BIT Word/Bit Word sequence.

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3.4.6.2.5.6 Scope of Return BIT Word Tests - The Return BIT Word shall not be required to perform tests on Unit 1. Unit 2 and Unit 3 functions are required to be tested by the Return BIT Word tests.

3.4.6.2.5.7 BIT Test Effectiveness - The equipment and Return BIT Words shall be so designed, that when malfunctions/failures occur, fault location, without ambiguity is determined to the WRU level. Such requirement shall apply at a rate of 95 correct fault locations for every 100 failure cases. Demonstration of this requirement shall be in accordance with paragraph 4.2.2.

3.4.6.2.5.8 BIT Multiple Word Analysis - Equipment and test words may be constructed such that failure determination and location is by computer analysis of several words, i.e., each word sequence does not have to be a complete test in itself.

3.4.6.3 Input Words - There shall be two types of input words to the computer from the AN/ASA-76 as follows:

(1) Status Word(s)
(2) BIT Words

3.4.6.3.1 Status Word - The Status Word is an input word to the Computer which is formatted by the equipment and generated as the normal response to a Command Word or Return Status Word.

3.4.6.3.1.1 Status Word Transfer Rates - The status word shall be input to the Computer in a time not to exceed 0.25 msec from receipt of a Return Status Word or Command Word by the AN/ASA-76.

3.4.6.3.1.2 Command in Progress - If a Computer generated command is received by the equipment while execution of a previously received command is in progress, the command in progress shall be continued to its normal completion. The newly received command shall be ignored except that the equipment will return a status word of command in progress along with an indication that the last Command Word was not processed. This paragraph shall not apply to the INSTANTANEOUS ABORT OF TRANSMISSION COMMAND.

3.4.6.3.1.3 Status Word (Manual Mode) - The Status Word shall contain the status of the modes and functions of the Unit 1 switches when the equipment is operated in the Manual Mode.

3.4.6.3.1.4 Status Word (Computer Mode) - The Status Word shall contain the modes/functions and RF channels of the last transmitted command or of the command in progress when Return Status Word follows a Command Word and the equipment is in the COMPUTER MODE.

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3.4.6.3.1.5 Status of BIT Word - When a Return Status Word follows a Return Bit Word, the Status Word shall contain the results of the last test performed by the Return BIT Word. In effect, the Status Word shall be input to the Computer in a time not to exceed 0.25 msec from receipt of the Return Status Word by the AN/ASA-76.

3.4.6.3.2 BIT Word - The BIT Word is an input word to the Computer which is formatted by the equipment and is the normal response to a Return BIT Word.

3.4.6.3.2.1 BIT Word Test Results - There shall be a series of BIT Word(s) each corresponding to particular tests generated by the Return BIT Word tests. Each BIT Word shall contain a code for identification of the test performed and/or the functions tested. Each BIT Word shall be returned to the Computer within 3.0 sec from the receipt of a Return BIT Word. For each function tests, the BIT Word shall contain a Go/No-Go indication.

3.4.6.4 Initial Conditions - When power is applied, CASS shall assume the following INITIAL MODES of operation:

(1) COMPUTER MODE
(2) SONIC PING - CW 1.0 sec for both Sensor Stations
(3) Sonar Channel A to Sensor Station I Upper Display
Sonar Channel B to Sensor Station I Lower Display
Sonar Channel C to Sensor Station II Upper Display
Sonar Channel D to Sensor Station II Lower Display
(4) Notch Filters, OUT
(5) RF NO CHANNEL CODE
(6) Display/Processor Modes to CASS

These modes shall be referred to as the INITIAL MODES.

3.4.6.4.1 Power Interrupt - Whenever a power interrupt is experienced, the equipment shall recover to its INITIAL MODE.

3.4.6.4.2 Transfer to Computer Mode - When control is transferred from one of the manual modes to COMPUTER MODE, the equipment shall assume its INITIAL MODES.

3.4.6.4.3 Transfer to Manual Mode - When the AN/ASA-76 control is transferred from COMPUTER MODE to a manual mode, the equipment shall not assume an automatic 16 or 32 second mode. It shall also not assume of VHF, Depth, Scuttle, or External mode.

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3.4.6.5 Interrupt Capability - The equipment shall have the capability of providing a pulse to the DIU for purposes of generating an External Interrupt to the Computer. This pulse shall be referred to as the INTERRUPT PULSE.

3.4.6.5.1 Interrupt Pulse Connector - The INTERRUPT PULSE shall be made available to the DIU on the 24J5 connector of Unit 2.

3.4.6.5.2 Interrupt Pulse Conditions - The INTERRUPT PULSE shall be generated by the equipment under the following conditions:

(1) A power interrupt has occurred and the equipment has initialized.

(2) A control mode change has taken place, i.e., computer to Q-47, Q-47 to one of the manual CASS modes, etc.

(3) When the equipment has emitted a command signal in one of the manual modes.

(4) When Manual Processor/Display reassignment has occurred.

3.4.6.5.3 Interrupt Status Word - The equipment shall be so designed that in INTERRUPT STATUS WORD is transferred to the DIU whenever an INTERRUPT PULSE is generated.

3.4.6.5.4 Interrupt Status Word (Manual Mode) - The INTERRUPT STATUS WORD shall contain all modes and functions of the Unit 1 switches when the equipment is operated in one of the manual modes.

3.4.6.5.5 Interrupt Status Word (Computer Mode) - The INTERRUPT STATUS WORD shall contain the initial conditions when COMPUTER MODE has been selected.

3.4.6.6 Manual Operation - The equipment shall be so designed that it may be disconnected from the DIU and retain normal manual modes of operation.

3.4.6.7 Order of Bits - When Computer Output Words are received by the AN/ASA-76 via the DIU and this specification requires that they be returned to the DIU, the word returned to the DIU shall be returned such that all bits can be transferred to the computer by the DIU with the same bit structure as received, i.e. bit "0" returned shall be in the same position when returned as bit "0" received.

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3.4.7 AN/AQA-9 Processor Control - The equipment shall supply the AN/AQA-9 a 30 bit serial computer word for control. This word is the same word received by the equipment from the digital interface unit. In addition, the functions selected by unit 1 must be coded in the same computer word format and supplied to the AN/AQA-9 as in computer control.

3.4.8 End to End BITE Test - In addition to the BUILT-IN-TEST required for the equipment, as specified, BIT shall also provide simulated CW, FM and AN/SSQ-47 targets with the sonic gates and mode control lines to the AN/AQA-7(V) to complete the end-to-end BITE tests on the AN/AQA-7(V) in the active mode. Identical end-to-end test signals shall be initiated from either unit 1 or the computer at the discretion of the operator. In performing the end-to-end tests the input frequency converters for all sonar channels (A, B, C, and D) shall be utilized during each test. The simulated targets, sonic gates, and mode controls shall be supplied to the AN/AQA-7(V) system such that all four displays for the AN/AQA-7(V) will present the same information.

3.4.8.1 CW Modes - The equipment shall supply CW test signals at 650, 700, 750, 800, 850, 900, and 950 Hz for all ping lengths. The tones shall end at 2, 4, 6, 8, 10, 12 and 14 seconds after the trailing edge of the sonic gate. The tones shall be mixed with noise to alternately produce 0 db signal-to-noise ratio and a minimum detectable signal.

3.4.8.2 FM Modes - The equipment shall supply FM UP and FM DOWN test signals. The FM slides shall end at 2, 4, 6, 8, 10, 12, and 14 seconds after the trailing of the sonic gate. The FM slides shall be mixed with noise to produce a 0 db signal-to-noise ratio followed by a minimum detectable signal.

3.4.8.3 AN/SSQ-47 Modes - The equipment shall supply AN/SSQ-47 test signals at 650, 700, 750, 850, 900 and 950 Hz for 0.1 seconds. The tones shall occur at 2, 4, 6, 8, 10, 12, and 14 seconds after the trailing edge of the sonic gate. The tones shall be mixed with noise to alternately produce a 0 db signal-to-noise ratio and a minimum detectable signal.

3.4.9 External Mode - Provisions shall be included to augment external sonic tones following address tones as specified in table IIIA and IIIB. (Note: Electrical Connection requirements per paragraphs 3.5.1.6, 3.5.2.6, and 3.5.3.6 below may be modified to provide this function.)

3.4.9.1 External Mode Operation - When the external mode function is selected the equipment shall generate addresses for either the active or passive sonobuoys and after a 2.5 milliseconds delay the equipment sends a sonic gate 3000 milliseconds long to the external source and during this time the equipment will accept signals from the external source to modulate the transmitter. This external signal shall be within the range of 5 kHz to 50 kHz and meet the requirements of the modulated signal as previously specified.

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3.5 Detail Requirements -

3.5.1 Control, Reference Signal Generator Unit 1 - The control, reference signal generator shall meet the applicable requirements of Specification MIL-C-6781 for control panels to the extent specified herein.

3.5.1.1 Functions - The control, reference signal generator shall contain the controls and circuits necessary for operation of the equipment in accordance with the requirements of this specification.

3.5.1.2 Form Factor - The control, reference signal generator, unit 1, form factor shall be in accordance with figure 1. The maximum height shall be no greater than 10.88 inches.

3.5.1.3 Weight - The weight of the control, reference signal generator shall not exceed 12 pounds.

3.5.1.4 Contents - The control, reference signal generator shall contain the following circuits:

(a) Address Code Generator
(b) Command-Sonic and Mode Select
(c) Start-Programmer
(d) Regulator Assembly
(e) Display and Notch Filter Switch
(f) BITE

3.5.1.4.1 Control Switch Circuits - The control switch circuits shall fulfill the functional requirements specified in 3.4.6. Manually programmed instructions shall be indicated through amber illumination of the activated function. Non-activated functions shall not be illuminated.

3.5.1.4.2 Control Logic Circuits - The control logic circuits shall control energization of the switch lamps illuminating the selected functions. An interlock shall be provided to prevent accidental initiation of depth and scuttle commands. Timing circuitry for automatic sequence shall be provided.

3.5.1.4.3 Control Encoder Circuits - The control encoder circuits shall generate coded control signals needed for generation of address, sonic or command tones.

3.5.1.4.4 Power Regulators - Voltage regulators shall be provided to supply power to the circuitry.

3.5.1.5 Controls - The controls shall be contained on a plastic lighted plate in accordance with MIL-T-7788, class 2-W, type V.

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3.5.1.6 Electrical Connections - Connections to external circuits shall be in accordance with EI-555, Avionics Installation Instruction for Sonar Computer-Recorder Group AN/AQA-7(V).

3.5.2 Reference Signal Generator Unit 2 - The reference signal generator shall meet the following requirements:

3.5.2.1 Function - The reference signal generator shall:

(a) Generate address, command sonic tones in accordance with 3.4.3.

(b) Process AN/SSQ-47 signals in accordance with 3.4.4.2.

(c) Process AN/SSQ-50 signals in accordance with 3.4.4.1.

(d) Process AN/SSQ-62 signals in accordance with 3.4.4.3.

3.5.2.2 Form Factor - The Reference Signal Generator Unit 2 form factor and mounting shall be in accordance with figure 2.

3.5.2.3 Weight - The weight of the Reference Signal Generator shall not exceed 33 pounds exclusive of mount.

3.5.2.4 Contents - The Reference Signal Generator shall contain the following assemblies, subassemblies, etc.

(a) Regulator
(b) Generator, Signal Tone
(c) Converter, Input Frequency A and B
(d) Converter, Input Frequency C and D
(e) Sync Signals
(f) Heterodyned Signals
(g) Computer Interface
(h) Keying Pulse
(i) BITE

3.5.2.5 Controls - The Reference Signal Generator shall contain no operational controls.

3.5.2.6 Electrical Connections - Connections to external circuits shall be in accordance with EI555, Avionics Installation instruction for Sonar Computer-Recorder Group AN/AQA-7(V).

3.5.3 Radio Transmitter Unit 3 - The Radio Transmitter shall generate and modulate a Radio Frequency carrier with signals, from the Reference Signal Generator, in accordance with 3.4.2.

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3.5.3.1 Functions - The Radio Transmitter shall contain the circuits and power supply required to generate and Amplitude Modulate Radio Frequency Signal. Provisions shall be included to operate in two modes, transmit and standby. Transmission shall occur only in the transmit mode.

3.5.3.2 Form Factor - The Radio Transmitter shall be in accordance with figure 3.

3.5.3.3 Weight - The weight of the Radio Transmitter shall not exceed 15 pounds, exclusive of mounts.

3.5.3.4 Contents - The Radio Transmitter shall contain the following module, assemblies, and subassemblies.

(a) Power Supply, AC/DC
(b) Carrier Generator, modulator, and RF Amplifier
(c) Modulator Control
(d) BITE

3.5.3.5 Controls - The Radio Transmitter shall contain no operational controls.

3.5.3.6 Electrical Connections - Connections to external circuits shall be in accordance with EI555 Avionics Installation Instructions for Sonar Computer-Recorder Group AN/AQA-7(V). The impedance of interface signals shall be as specified in 3.4.1.2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in a contract or purchase order, the supplier is responsible for the performance of all test requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the tests set forth in the specification where such tests are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements.

(a) Preproduction (First Article) Tests
(b) Initial Production Tests
(c) Acceptance Tests
(d) Life Tests

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4.2 Preproduction (First Article) Tests - Preproduction (First Article) tests shall be made by the contractor on an equipment representative of the production equipments to be supplied under the contract. Preproduction (First Article) tests shall be accomplished under the approved test procedure of 4.6. The Government Inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction (First Article) Test Data - The contractor shall submit all data collected in conducting these tests to the procuring activity for review.

4.2.2 Scope of Tests - Preproduction (First Article) tests shall include all tests deemed necessary by the procuring activity to determine whether or not the equipment meets all the requirements of this specification and the contractor. Preproduction (First Article) tests shall include environmental tests in accordance with the procedures of Specification MIL-T-5422 interference tests in accordance with Specification WR101 Part 1, MIL STD 461 and Maintainability Tests in accordance with the MIL STD 470 and as follows:

4.2.2.1 Maintainability Assurance Test - It shall be the specific objective of the demonstration to determine the maximum time to perform organizational level corrective maintenance action, ($M_{Max_{ct}}$) as per paragraph 3.7.

4.2.2.2 Demonstration Methods - The demonstration shall utilize both manually operated BIT and computer SYGNOG and Diagnostic programs. The demonstration shall be conducted at the contractor's facility when manual modes are demonstrated and at the Weapon Systems Integrations Facility when computer test modes are demonstrated. The government shall supply additional avionics equipment required to perform the tests.

4.2.2.3 Demonstration Technician - Navy enlisted technicians will be temporarily assigned to the test site for the express purpose of conducting the demonstration. Prior to the demonstration, the technician will have attended a factory maintenance training course conducted by the manufacturer.

4.2.2.4 Failure Selection - The contractor shall provide to the procuring activity a list of 300 failures from which 75 failures for demonstration will be selected. Each of the 300 failures submitted will be different and independent from any of the others. Each will represent typical or likely failures that may occur to the CASS equipment. The distribution of the 300 failures selected shall be such that each mode and function of the CASS shall be subjected to the maintenance demonstration. The density of distribution

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shall be based on predicted failure rates. For each failure submitted, the contractor shall supply, as a minimum, the following data:

(1) Nomenclature of part failed
(2) Type failure - short, open, etc.
(3) Expected symptoms
(4) Details of test designed to detect fault
(5) Details of test designed to isolate fault
(6) Predicted failure rate
(7) The designed time to complete the maintenance action

4.2.2.5 Pass/Fail Criteria - If all maintenance actions are successfully completed in 30 minutes, that event shall be considered as PASSED. If at the end of 30 minutes there are still maintenance actions to be performed, that event will be considered as FAILED. If more than 5 maintenance events are FAILED during the course of the (M_{max}) demonstration, it shall be considered as failure to demonstrate specified maintainability.

4.2.3 Preproduction (First Article) Approval) - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipment shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 Production Requirements - Equipment supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction (first article) sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction (first article) sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. No other tests shall be conducted on equipment prior to starting the initial production tests. The preproduction sample shall not be selected for this test.

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4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for Initial Production Tests the contractor shall also submit such accessory material and data necessary to the test equipment.

4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during these tests shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests may be cause for deliveries of equipment under the contract to cease until proper corrective action is approved by the procuring activity and accomplished by the contractor. Corrective action shall also be accomplished on equipment previously accepted by the Government when requested by the procuring activity.

4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test, the equipment shall be reworked by the contractor by replacing all worn or damaged items. After reworking the contractor may resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitive results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(a) Individual Tests
(b) Sampling Tests
(c) Reliability Assurance Tests
(d) Special Tests

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4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(a) Examination of Product
(b) Operational Test
(c) Manufacturing Run in Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operations Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Run in Test - Each equipment shall be operated under the condition specified herein for a period of ten hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment. This test shall be deleted if the reliability test includes a test on each equipment which consumes at least ten hours of operation.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ±3 g's

The equipment shall be vibrated (without vibration isolators) for a period of ten minutes prior to beginning of the ten hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for five minutes and lateral to the plane for five minutes. Where it is not feasible to vibrate the equipment in two directions the vertical direction shall be used. During the ten hour period of operation following the ten minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the ten minute vibration period need not be repeated when it is certain

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the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The ten hour period specified above may be composed of two 5 hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the Government inspector in accordance with the following:

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Sampling Test
First 10	1
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

One of the first ten need not be selected and tested if initial production tests are conducted. Sampling Tests are not required when Reliability Assurance Tests are conducted.

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual and special tests.

(3) Manufacturing run in test specified in 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

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4.4.3 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted using MIL-STD-781. Tests as required by both the Qualification Phase and the Sampling Phase shall be conducted. Classification of failure shall be in accordance with MIL-STD-781 and AR-34.

4.4.3.1 Qualification Phase - Prior to the acceptance of equipments under the contract or order, a minimum of three equipments shall be tested as outlined in MIL-STD-781, under a section entitled "Qualification Phase of Production Reliability Tests." The maximum number of equipments to be used shall be those listed in table 5 of MIL-STD-781. For the Qualification Phase, Test Level E shall be used. The Accept-Reject Criteria for Test Plan 5 shall be used. The following exceptions to Level E and MIL-STD-781 shall be allowed:

(a) The operating temperature shall be -30° to +55°C.

4.4.3.2 Reliability Production Acceptance Phase Test - The equipment, throughout production shall be tested as outlined in MIL-STD-781 (as modified herein), under the Section entitled "Sampling Phase of Production Reliability Test." For the Sampling Phase (or All-Equipment Test), Test Level E shall be used.

4.4.3.2.1 All Equipment Test - Each equipment produced except those submitted for the Reliability Qualification Test, shall be tested for 50 hours. Prior to the 50 hour test on each equipment, a burn-in period may be used at the option of the contractor. If the burn-in period is to be used the details thereof must be included in the approved test procedures. To determine whether the MTBF is being met at any time during the contract the operating test hours and the failures thereon (not counting burn-in failures or burn-in operating time) shall be totaled and the results compared with the reject line of Test Plan II of MIL-STD-781. (Extend the line as necessary to accommodate the data.) These totals shall accumulate so that at any time the experience from the beginning of the contract is included. At the conclusion of each month the test results shall be sent to the procuring activity and to the Naval Air Systems Command, Attention: Avionics Division. At any time that the current totals of test hours and test failures plotted on Test Plan II curves show a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of equipment at any time that a reject situation exists pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc., so that the entire compilation will show other than a reject decision. The Radio Transmitter, may, at the manufacturer's option be subjected to the requirements of this paragraph independent of the balance of the system.

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4.4.3.3 Test Details - The test details such as the length of the test cycle, the length of heat portion of the cycle, the performance characteristics to be measured, special feature criteria, preventive maintenance to be allowed during the test, etc., shall be part of the test procedures to be submitted and approved by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests.

4.4.4 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(a) On an early equipment after an engineering or material change.

(b) Whenever failure reports or other information indicate that additional tests are required. (This will be determined by the procuring activity.)

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction (first article) test shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.4.3 Equipment Failure - Should a failure occur during either the sampling, reliability assurance, special tests or maintainability assurance tests, the following steps shall be taken:

(a) Determine the cause of failure.

(b) Determine if the failure is an isolated case or design defect.

(c) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(d) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

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4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life tests. The test shall be of 300 hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the Government inspector in accordance with the following. (Equipments which have successfully passed the Initial Production Test, Sampling Tests, Reliability Tests, or Special Tests may be selected for life tests.) When reliability tests are conducted, the life test shall be omitted if, during the reliability tests, a quantity of equipments equal to, or more than, that listed below receive at least 300 hours each of test time.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal room
Altitude	Normal ground (0 - 5000 ft)
Humidity	Room ambient
AC Voltage	115 ±5 volts (at lowest applicable frequency)
DC Voltage	27.5 ±2.0 volts

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8 hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.6 Test Procedures - The procedures used for conducting preproduction (First Article) and acceptance tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government Inspector to modify the

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tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to Preproduction (first article), Initial Production or Acceptance Tests may be reconditioned by the contractor by replacing all worn or damaged items. After reworking, the contractor may resubmit the equipment for acceptance.

4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the equipment, shall be furnished the Government Inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in specification MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

6. NOTES

6.1 Intended Use - The equipment produced in accordance with this specification shall complete the requirements for the addition of a Command Activated Sono function to the present functional capability of the AN/AQA-7(V). The equipment shall be capable of generating the transmitting activating commands and preprocessing the returned data to the extent specified herein.

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6.2 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Selection of applicable levels of packaging and packing (see 5.1)

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(a) Contract - The contract shall have precedence over any specification.

(b) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable shall be specifically approved in writing by the procuring activity.

(c) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.5 Non-Repairable Subassemblies - As a general rule, non-repairable subassemblies should be encapsulated or hermetically sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and rating should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should demonstrate a Mean-Time to Failure greater than 5,000 hours, and for many applications this figure must be nearer 50,000 hours.

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6.6 Type Designations - The parentheses when used in the equipment type designation will be deleted or replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.8. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.7 Associated Equipment - The equipment shall operate with the following equipments:

(a) Recorder AN/AQH-4
(b) Aircraft Data Processor AN/AQA-7(V) - MIL-A-81486(AS)
(c) Sonar Display Computer AN/AQA-9 - XAS-1645(AV)
(d) Sonobuoy AN/SSQ-47 - MIL-S-81349
(e) Sonobuoy AN/SSQ-50 - MIL-S-81666
(f) Sonobuoy AN/SSQ-62 - XAS-1647(V)
(g) Radio Receiving Set AN/ARR-72
(h) Transceiver-UHF-2 MIL-R-81628
(i) Radio Receiving Set AN/ARR-75 - MIL-R-81681

6.8 FM Linearity Measurement - The FM sweep may be stopped at selected intervals to measure frequency deviations from a linear function.

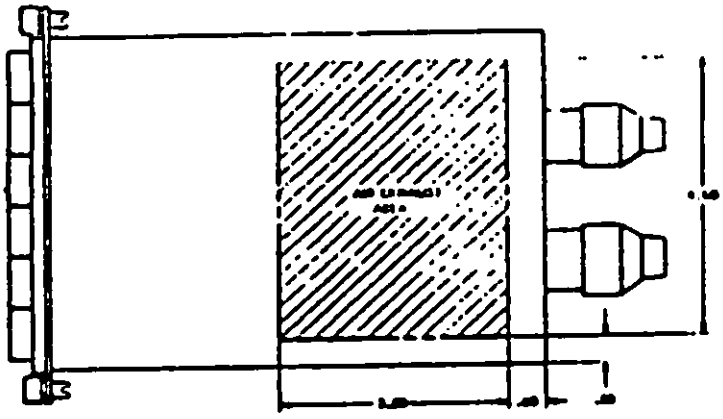
6.9 Lamp Test Operation - The lamp test of Unit 1 is performed by the sonar operator using the lamp test switch on the AN/AQA-7(V) Control Panel. The operation of this switch is as follows:

(a) When the switch is in position 1 the lamps of the multi-lamp push buttons are lit in the diagonal line from the left hand corner to the lower right hand corner of each switch assembly and all of the single push buttons to the left of the center line of the unit.

(b) When the switch is in position 2 the lamps of the multi-lamp push buttons are lit in the diagonal line from the right hand corner to the lower left hand buttons to the right of the center line of the unit.

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NOTES

1. [illegible]
2. [illegible]
   IEC - [illegible]
   ISO - [illegible]
3. [illegible]
4. ALL DIMENSIONS ARE [illegible]
5. [illegible]
6. [illegible]
7. ADD 3.3 INCHES TO THE PLUG REMOVAL CLEARANCE DIMENSION FOR THE BENDING RADIUS CLEARANCE REQUIRED PER CABLE
8. [illegible]
9. [illegible]

CONNECTOR TABLE	
REF SYMBOL	MILITARY PART NUMBER
[illegible]	[illegible]
[illegible]	[illegible]

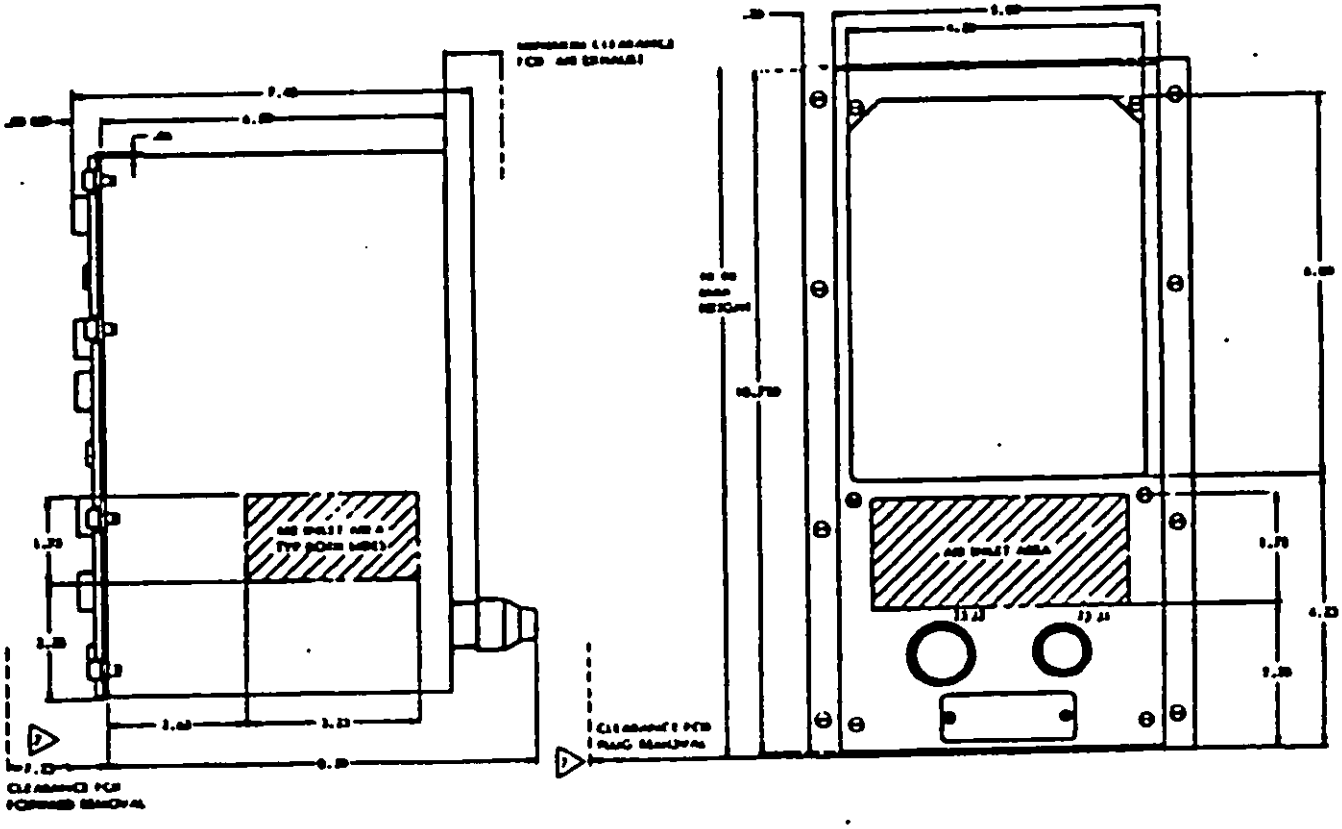
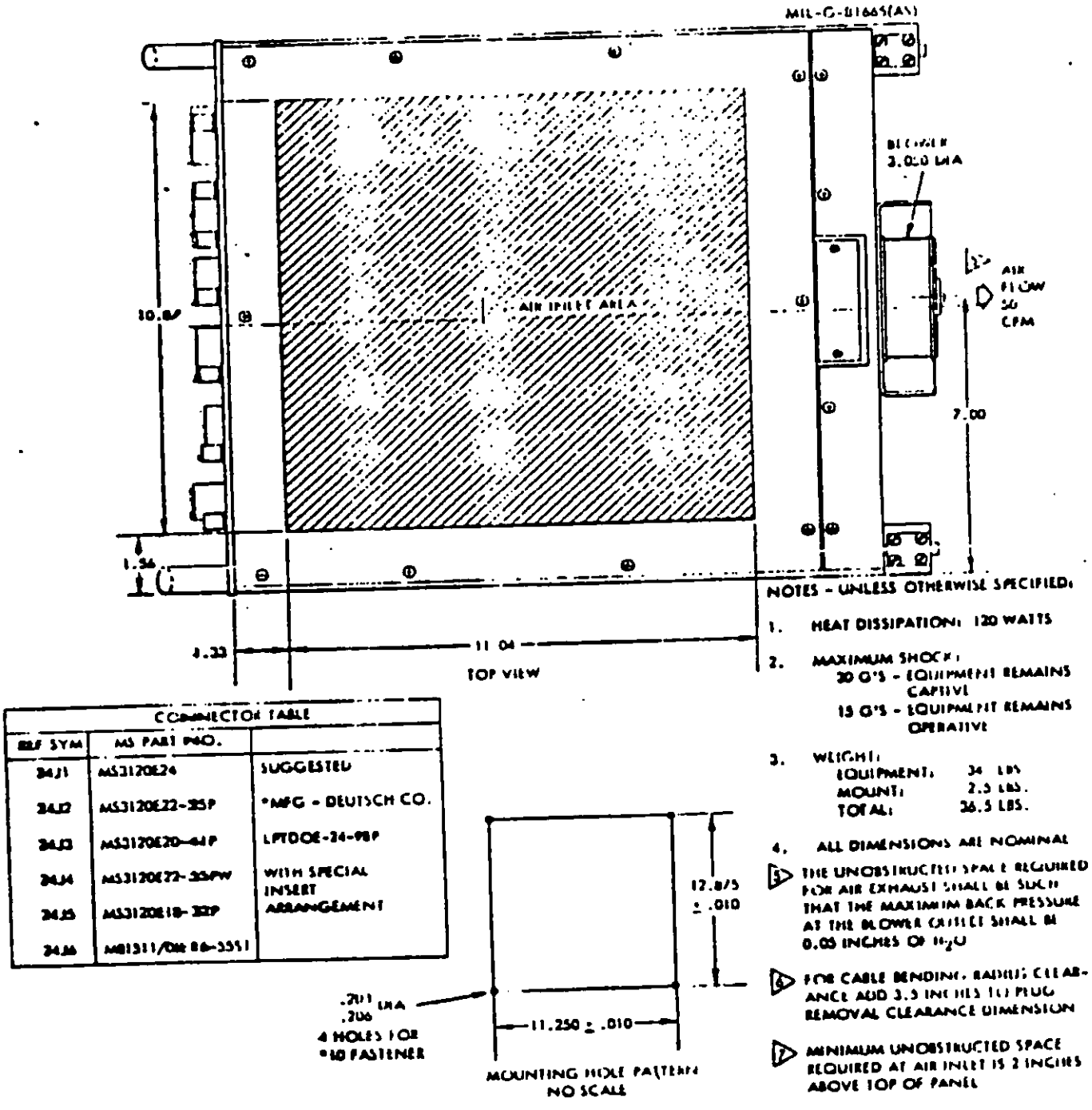


FIGURE 1. UNIT 1 AN/ASA-76 CONTROL, REFERENCE SIGNAL GENERATOR AND OUTLINE DIMENSIONS

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CONNECTOR TABLE		
REF SYM	MS PART NO.	
J4J1	MS3120E24	SUGGESTED
J4J2	MS3120E22-35P	*MFG - DEUTSCH CO.
J4J3	MS3120E20-41P	LPTDOE-24-98P
J4J4	MS3120E22-55PW	WITH SPECIAL INSERT ARRANGEMENT
J4J5	MS3120E18-32P	
J4J6	MB1311/0[illegible]-3551	

FIGURE 2-1 UNIT 2 AN/ASA-76 REFERENCE SIGNAL GENERATOR AND MOUNTING OUTLINE DIMENSIONS

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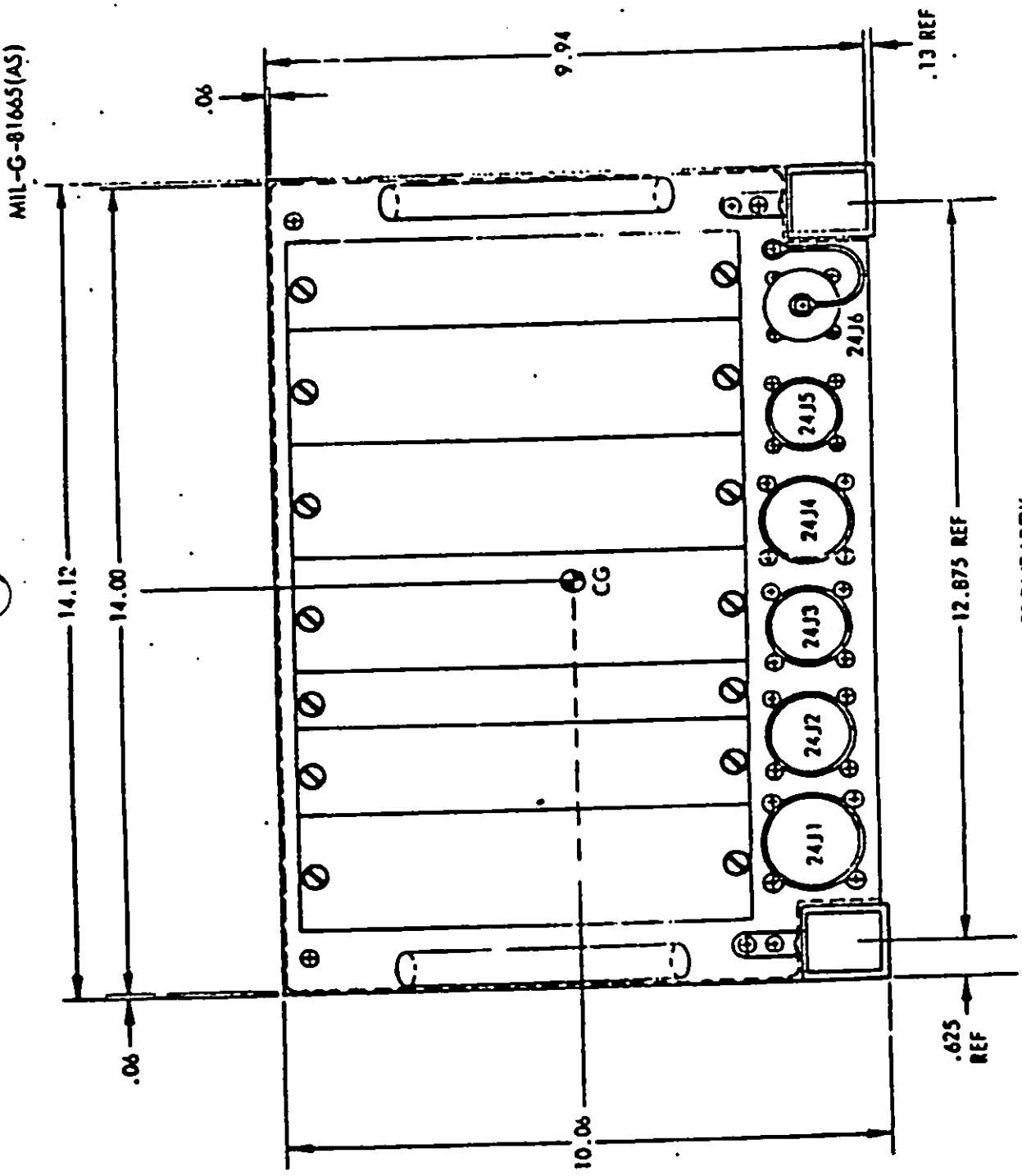


FIGURE 2-2 UNIT 2 AN/ASA-76 REFERENCE SIGNAL GENERATOR AND MOUNTING OUTLINE DIMENSIONS

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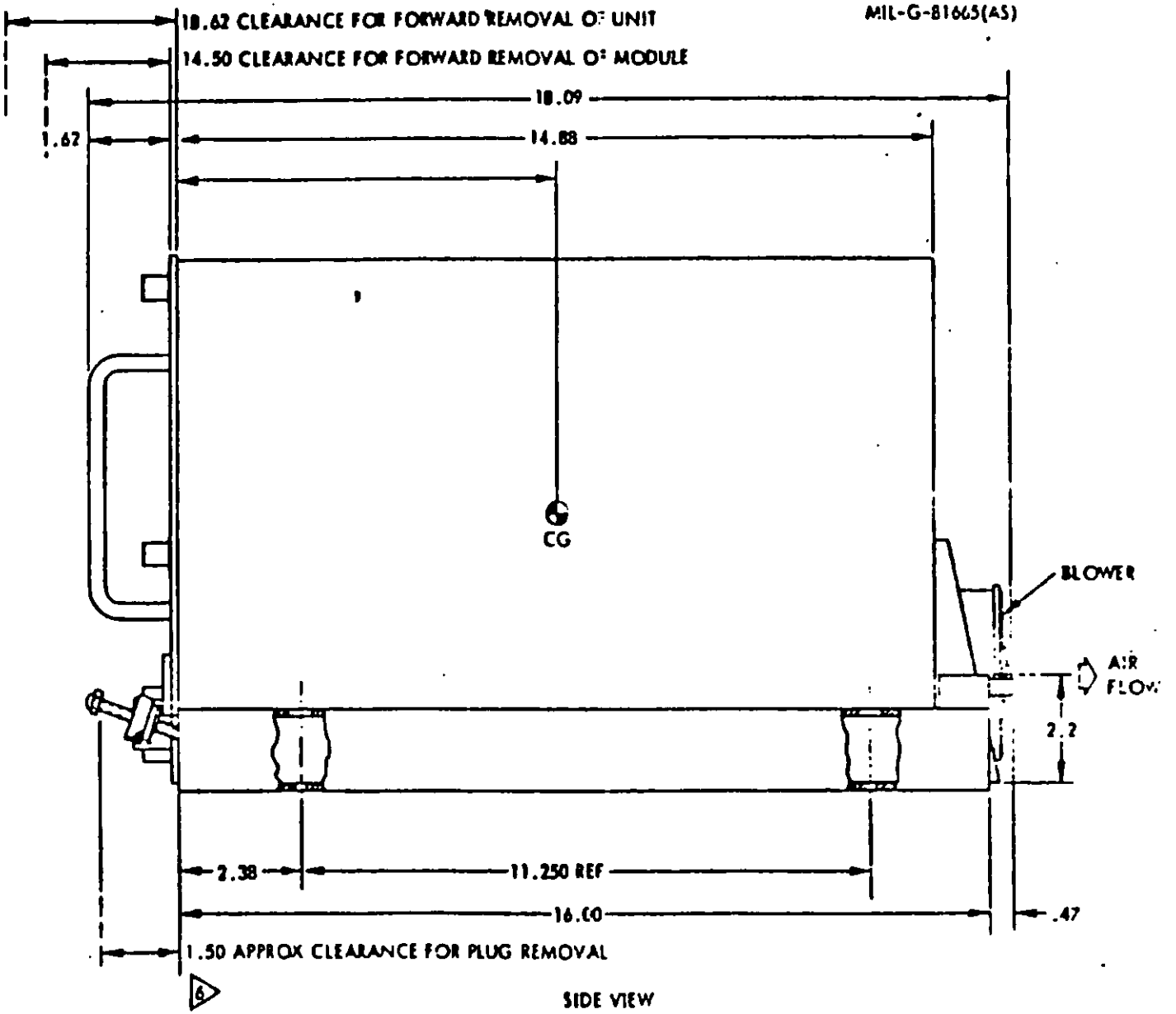
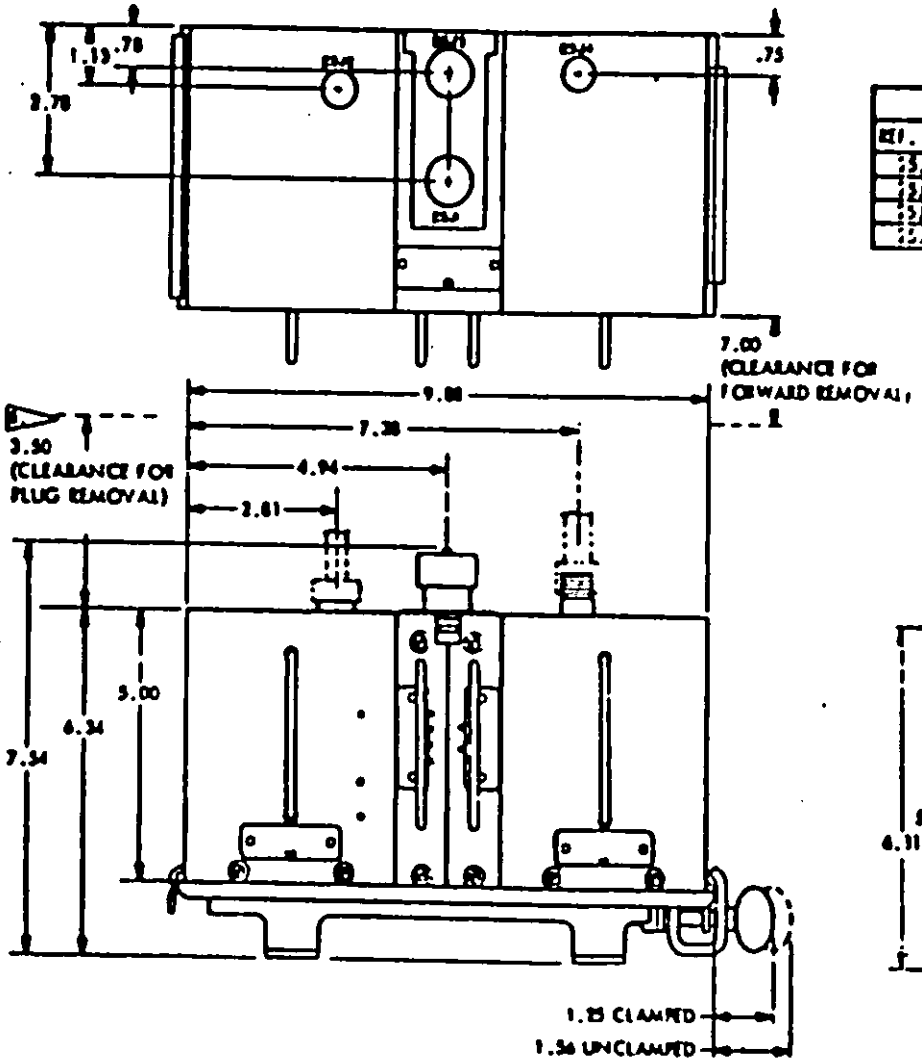


FIGURE 2-3 UNIT 2 AN/ASA-70 REFERENCE SIGNAL GENERATOR AND MOUNTING OUTLINE DIMENSIONS

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CONNECTOR TABLE	
REF. SYM	MS PART NO.
J1	MS3511/21E16-35S1
J2	MS3120E16-6P
J3	MS3[illegible]16-1S1
J4	UG [illegible] A/U

NOTES: UNLES OTHERWISE SPECIFIED

1. HEAT DISSIPATION: 80.0 WATTS
2. MAXIMUM SHOCK:
   20 G'S - EQUIPMENT REMAINS CAPTIVE
   15 G'S - EQUIPMENT REMAINS OPERATIVE
3. WEIGHT
   EQUIPMENT 13.0 LBS
   MOUNT 3.0 LBS
   TOTAL 17.0 LBS
4. ALL DIMENSIONS ARE NOMINAL
5. THERMAL CONTACT METHOD-FREE CONNECTION

▷ FOR CABLE BENDING RADIUS CLEARANCE ADD 3.5 INCHES TO PLUG REMOVAL CLEARANCE DIMENSION.

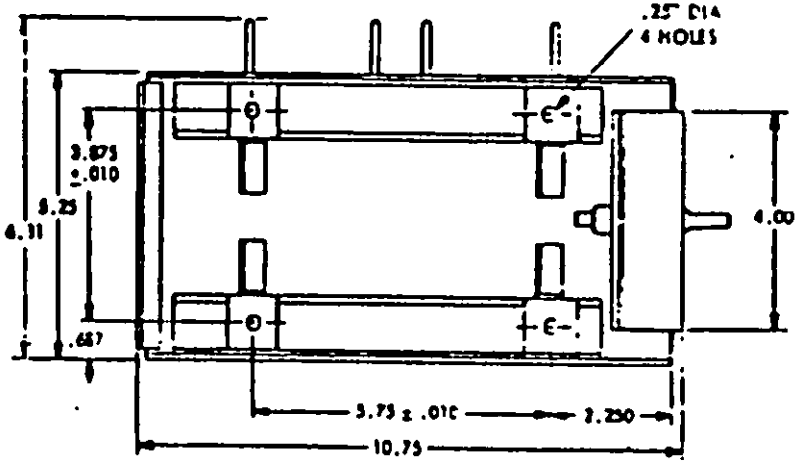


FIGURE 3 UNIT 3 AN/ASA-76 RADIO TRANSMITTER AND MOUNTING OUTLINE DIMENSIONS

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